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RESPIRATOR APPARATUS

Abstract

One variation of a respirator apparatus includes a shield center section, a filter assembly, and a gasket. The shield center section defines: a transparent center region configured to align with a nose and a mouth of a user; and a filter window laterally offset from the transparent center region. The filter assembly includes a filter media and a seal: encircling a perimeter of the filter media; and defining a center groove configured to mate with a rim of the filter window to transiently retain the filter assembly within the filter window. The gasket is arranged on a rear face of the lens center section and configured to: seal against a face of the user; and locate the lens center section and the filter assembly offset from the face of the user to form an airflow channel to direct airflow across a glabrous cheek region of the user.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. Non-Provisional application Ser. No. 18/520,316, filed on 27 Nov. 2023, which claims the benefit of U.S. Provisional Application No. 63/428,314, filed on 28 Nov. 2022, each of which is hereby incorporated in its entirety by this reference.

TECHNICAL FIELD

[0002] This invention relates generally to the field of respirators and, more specifically, to a new and useful apparatus for a transparent shield respirator with conformal foam facial gasket incorporating interchangeable voice translucent, conformal frame pleated filters in the field of respirators.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0003] FIG. 1 is a schematic representation of a respirator apparatus;
[0004] FIG. 2 is a schematic representation of the respirator apparatus;
[0005] FIG. 3 is a schematic representation of the respirator apparatus;
[0006] FIG. 4 is a schematic representation of the respirator apparatus;
[0007] FIG. 5 is a schematic representation of the respirator apparatus;
[0008] FIG. 6 is a schematic representation of the respirator apparatus;
[0009] FIGS. 7A and 7B are schematic representations of the respirator apparatus;
[0010] FIG. 8 is a schematic representation of the respirator apparatus; and
[0011] FIG. 9 is a schematic representation of the respirator apparatus.

DESCRIPTION OF THE EMBODIMENTS

[0012] The following description of embodiments of the invention is not intended to limit the invention to these embodiments but rather to enable a person skilled in the art to make and use this invention. Variations, configurations, implementations, example implementations, and examples described herein are optional and are not exclusive to the variations, configurations, implementations, example implementations, and examples they describe. The invention described herein can include any and all permutations of these variations, configurations, implementations, example implementations, and examples.

1. Respirator Apparatus

[0013] As shown in FIG. 1, a respirator apparatus **100** includes: a shield center section **110**; a first filter assembly **120**; a second filter assembly **128**; and a gasket **140**.

[0014] The shield center section **110** is formed of a flexible material and is transparent in its entirety defining a center region **112** configured to locate centered over a face of a user; a left filter window **114** laterally offset from the transparent center region **112**; and a right filter window **116** laterally offset from the transparent center region **112** opposite the left filter window **114**.

[0015] The first filter assembly **120** includes a first filter media **121** and a first seal **122**. The first seal **122**: encircles a periphery of the first filter media **121**; and defines a first center groove **123** configured to mate with a first rim **115** of the left filter window **114** to transiently retain the first filter assembly **120** within the left filter window **114**.

[0016] The second filter assembly **128** includes a second filter media **129** and a second seal **130**.

The first seal **122**: encircles a periphery of the second filter media; and defines a second center groove **131** configured to mate with a second rim of the right filter window **116** to transiently retain the second filter assembly **128** within the right filter window **116**.

[0017] The gasket **140**: is arranged on a rear face of the shield center section **110**; and configured to conform with the contours of the face of the user and to center the transparent unobstructed window of the center region **112** with a nose and a mouth of the user. The gasket **140** is also configured to locate the shield center section **110** and the first filter assembly **120** offset from the face of the user to form a left airflow channel **150**: defining a first height; defining a first depth less than the first height; and extending across a left cheek region of the user exposing glabrous skin regions on the upper cheek. The gasket **140** is also configured to locate the shield center section **110** and the second filter assembly **128** offset the face of the user to form a right airflow channel **150**: defining a second height; defining a second depth less than the second height; and extending across a right cheek region of the user exposing glabrous skin regions on the upper cheek.

2. Applications

[0018] Generally, the respirator apparatus **100** can function as a respirator that: induces a cooling sensation across glabrous skin areas of the upper cheek region of the user; maintains contact with the face of the user by conforming responsively to movement of the face (e.g., head turning, speaking) by the user, demonstrating elasticity; and enables other users to view facial cues (e.g., expressions) from the user. More specifically, the respirator apparatus **100** can include: a shield center section **110** including a transparent region **112** configured to center over a nose and/or a mouth of the user; a filter assembly **120** transiently coupled to the shield center section **110** and arranged offset the transparent region **112** of the shield center section **110**; and a gasket **140** arranged on a rear side of the shield center section **110** and configured to conformally mate with the face of the user to locate the filter assembly **120** proximal the glabrous cheek region of the user and form an airflow channel **150** within an interior volume of the shield center section **110** between the filter assembly **120** and the transparent center region **112** of the shield center section **110** identified as the plenum.

[0019] Thus, during inhalation (e.g., nasal inhalation, oral inhalation) and exhalation (e.g., nasal exhalation, oral exhalation) by the user, airflow within the airflow channel **150** is directed across the glabrous skin on the upper cheek region to induce the cooling sensation across the face of the user while maintaining an unobstructed view of the nose and the mouth of the user within the transparent center region **112** of the shield center section **110**.

[0020] The respirator apparatus **100** includes the gasket **140**: formed of an elastomeric conformal closed cell foam material including a non-porous skin surface; arranged about a periphery of the shield center section **110**; including an upper gasket section **143** defining an arched geometry contoured along an upper edge of the transparent center region **112** of the shield center section **110**; and including a lower gasket section **144** defining a concave geometry contoured along a lower edge of the transparent center region **112** of the shield center section **110**. Additionally, the gasket **140** is configured to: conform against the face of the user to form a plenum airflow channel **150** extending between the filter assembly **120** and the transparent center region **112** within the interior of the shield center section **110**; contour along a nasal bone of the user to maintain the upper gasket section **143** in contact with the face of the user; and contour along the mandible of the user to maintain the lower gasket section **144** in contact with the face of the user.

[0021] The respirator apparatus **100** can further include: a neck harness **160** coupled to the shield center section **110** and configured to wrap about a rear neck of the user; and a cranial harness **165** coupled to the shield center section **110** and configured to form a halo configuration about the cranium of the user. Accordingly, the neck harness **160** cooperates with the cranial harness **165** to: stabilize the gasket **140** about the face of the user to form the airflow channel **150** within the shield center section **110**; locate the nose and the mouth of the user within the transparent center region **112** of the shield center section **110**; and locate the filter window **114**—and therefore the filter

assembly **120**—proximal the glabrous cheek region of the user.

[0022] Thus, the gasket **140** cooperates with the face of the user such that—rather than the gasket **140** sliding across the face of the user during movement of the face (e.g., head turns, mandible movement during speaking) by the user—the gasket **140** deflects, stretches, and compresses to conform to facial topology demonstrating homogenous elasticity, thereby: maintaining contact between the upper gasket section **143** and the nasal bone of the user; maintaining contact between the lower gasket section **144** and the bottom side of the mandible of the user; and directing airflow through the filter assembly within the interior volume of the shield center section **110**.

[0023] Additionally, the shield center section **110** further includes a filter window **114**: offset from the transparent center region **112** and configured to receive the filter assembly **120**; and configured to receive the filter assembly **120** to locate the airflow channel **150** proximal the glabrous cheek region of the user. In particular, the filter assembly **120** can include: a filter media **121** (e.g., polytetrafluoroethylene membrane, spunbonded polyester); and a seal **122** encircling the filter media **121** and defining a center groove **123** extending about the periphery of the seal **122** and configured to mate with a rim **115** of the filter window **114** to locate the filter assembly **120** within the filter window **114**. The seal **122** can: be formed of an elastomeric material configured to deform during installation of the filter assembly **120** into a filter window **114**; and include a groove configured to expand around and to seal against inner and outer edges of a rim **115** of the filter window **114**, thereby locating and sealing the filter assembly **120** within the filter window **114**. Accordingly, the filter assembly **120**: is operable in a deformed configuration by the user to locate the filter assembly **120** within the filter window **114** of the shield center section **110**; and is operable in a nominal configuration to locate the rim **115** of the filter window **114** within the center groove **123** of the seal **122** and thus form an interference fit between the filter window **114** and the filter assembly **120** that directs airflow from an exterior side of the shield center section **110** through the filter assembly **120** within the interior volume of the shield center section **110**. Thus, during inhalation by the user (e.g., from the nose, mouth), airflow within the airflow channel **150** is directed from the filter assembly **120** across the glabrous cheek region of the user to induce the cooling sensation. Additionally, during exhalation by the user (e.g., from the nose, mouth), airflow within the airflow channel **150** is directed from the transparent center region **112** across the glabrous cheek region of the user to induce the cooling sensation.

[0024] Therefore, the respirator apparatus **100**: induces a cooling effect across the glabrous cheek region of the user during inhalation and exhalation of the user; enables an outside observer to view facial cues from the nose and the mouth region of the user located within the transparent center region **112** of the shield center section **110**; deflects and twists about the face of the user to maintain the gasket **140** against the face of the user to direct airflow from the filter assembly **120** toward the interior volume of the shield center section **110**; and prevents contaminants (e.g., airborne particulates, dirt, dust) about an exterior environment of the shield center section **110** from entering the interior of the shield center section **110** identified as the plenum.

3. Lens Section

[0025] Generally, the respirator apparatus **100** includes a shield center section **110**: formed of an transparent flexible material (e.g., polycarbonate film) configured to deflect and twist responsive to movement of the face by the user; and configured to contour a face of the user to expose a nose and a mouth region of the user. More specifically, the respirator apparatus **100** includes: a transparent center region **112** configured to locate over a face (e.g., nose, mouth) of the user; a left filter window **114** laterally offset from the transparent center region **112**; and a right filter window **116** laterally offset from the transparent center region **112** opposite the left filter window **114**. Thus, the respirator apparatus **100**: exposes the nose region and the mouth region to an observer when arranged on the face of the user; and supports transient coupling of a filter assembly **120** at the left filter window **114** and/or the right filter window **116** to prevent contaminants (e.g., airborne particulates, dirt, dust) about an exterior environment of the lens center section **110** from entering

the interior of the shield center section **110**.

[0026] In one implementation, the shield center section **110**: defines a target width configured to curve from the left zygomatic bone to the right zygomatic bone of the user to locate the shield center section **110** to across a breathing zone (e.g., mouth, nose) and glabrous cheek regions of the user; and defines a target height configured to extend from a bridge of the nose to below a chin of the user. The shield center section **110** includes: an upper edge defining an upper arch configured to contour along a nose bone of the user; a lower edge defining a lower arch configured to contour along a mandible of the user; a left edge interposed between the upper edge and the lower edge and configured to contour along a left zygomatic bone of the user; and a right edge, opposite the left edge, interposed between the upper edge and the lower edge and configured to contour along a right zygomatic bone of the user. More specifically, the upper edge can further define: a left scalloped region configured to contour along a left maxilla bone, below a left eye, of the user to form an unobstructed field of view from the left eye when locating the lens center section **110** on the face of the user; and a right scalloped region—opposite the left scalloped region—configured to contour along a right maxilla bone, below the right eye, of the user to form an unobstructed field of view from the right eye when locating the shield center section **110** on the face of the user.

[0027] In this implementation, the shield center section **110** includes the left filter window **114**: arranged proximal the left side of the lens center section **110**; defining a target shape (e.g., semi-elliptical shape, circular shape, lozenge shape); defining a window width; and defining a window height less than the window width. Similarly, the shield center section **110** includes the right filter window **116**: arranged proximal the right side of the shield center section **110**; defining the target shape (e.g., semi-elliptical shape, circular shape, lozenge shape); defining the window width; and defining the window height less than the window width. Furthermore, the filter window **114** can include indicia (e.g., contrasting color band) arranged about the periphery of the filter window **114** representing a sealing edge configured to guide a user to locate the filter assembly **120** within the filter window **114** of the shield center section **110**.

[0028] Although this implementation includes a left filter window **114** and a right filter window **116** arranged on the shield center section **110**, other variations can include a single filter window and/or multiple (e.g., greater than two) filter windows arranged on the shield center section **110** configured to receive the filter assembly **120**.

[0029] Therefore, the shield center section **110**: enables observers to view facial cues (e.g., facial expressions) from the user through the transparent center region **112**; includes the filter assembly **120** transiently coupled within the right filter window **116** and the left filter window **114** of the shield center section **110**; and flexes and twists, responsive to movement of the face by the user.

4. Filter Assembly

[0030] Generally, the filter assembly **120** includes: a filter media **121**; and a seal **122**. The filter media **121**: is configured to prevent contaminants (e.g., airborne particulates, dirt, dust) from passing from an exterior environment outside of the lens center section **110** into the interior volume of the respirator apparatus **100** between the shield center section **110** and the user's face. The seal **122**: is arranged about (e.g., overmolded) the filter media **121**; and defines an outer circumferential groove configured to seat around and compress against the rim **115** of a filter window **114** to thus form a double-seal against the inner and outer edges of the rim **115** of the filter window **114** while locating the filter media **121** within the filter window **114**. More specifically, the filter assembly **120** is configured to deform-responsive to an applied force about the seal **122** encircling the filter media **121**—to locate the filter assembly **120** entirely within the filter window **114**. Accordingly, upon release of the applied force about the seal **122**, the filter assembly **120** returns to a nominal shape (e.g., semi-elliptical shape) to induce mating between the center groove **123** arranged about the seal **122** and the rim **115** about the filter window **114**, thereby forming the interference fit between the filter window **114** and the filter assembly **120**. Thus, the filter assembly **120** can: block contaminants (e.g., airborne particulates, dirt, dust) from entering an interior of the shield center

section **110**; and form an interference fit between the filter assembly **120** and the filter window **114** to direct airflow from outside the shield center section **110** through the filter assembly **120** into the interior volume of the shield center section **110**.

4.1 Interference Fit

[0031] In one implementation, the respirator apparatus **100** includes: the filter window **114** (e.g., left filter window **114** and/or right filter window **114**) defining a square rim **115** about a periphery of the filter window **114**; and the center groove **123** defining a forked geometry extending about a periphery of the seal **122** encircling the filter media **121** and configured to mate with the square rim **115** of the filter window **114**.

[0032] In one example, the forked geometry includes: a first tine **124** encircling the periphery of the seal **122**; a second tine **125**—opposite the first tine **124**—encircling the periphery of the seal **122**; and a slot **126** interposed between the first tine **124** and the second tine **125** configured to receive the square rim **115** of the filter window **114**. Accordingly, the forked geometry of the center groove **123** extends about the periphery of the seal **122** to mate with interior and exterior edges of the rim (e.g., four-sided rim, triangular rim, circular rim) about the periphery of the filter window **114** to form the interference fit between the filter assembly **120** and the shield center section **110**. In this example, the center groove **123** can transiently couple the filter window **114** to: locate the square rim **115** of the filter window **114** within the slot **126** of the forked geometry of the center groove **123**; seal the first tine **124** of the forked geometry against exterior edges of the rim **115** of the filter window **114**; and seal the second tine **125** of the forked geometry against interior edges, opposite the exterior edges, of the rim **115** of the filter window **114**, thereby forming the interference fit between the filter assembly **120** and the filter window **114**.

[0033] In one implementation, the filter assembly **120** is operable: during application of a force about the periphery of the filter assembly **120**, in a deformed configuration to locate the filter assembly **120** within the filter window **114**; and, during absence of an applied force about the periphery of the filter assembly **120**, in a nominal configuration to seal the filter assembly **120** against the rim of the filter window **114**.

[0034] In one example, the filter media **121**: includes a multi-layer construction (e.g., single-layer or two-layer polytetrafluoroethylene membrane, where the polytetrafluoroethylene membrane layer is bonded on each side to spunbonded polyester scrim) of a target filtering efficiency (e.g., between 95% for R95, N95 filtering efficiency and 99.97% for P100, N100 filtering efficiency); defines a parallel pleated geometry enclosed within the seal **122** about the filter media **121**; and spans a target surface area (e.g., between 80 square inches and 110 square inches, between 51000 square millimeters and 70967.6 square millimeters) within the seal **122**. Additionally, the seal **122**: is formed of an elastomeric material (e.g., urethane, thermoplastic) overmolded about the filter media **121**; and defines a target durometer range (e.g., between 55 Shore A and 75 Shore A) configured to deform the filter assembly **120** responsive to an applied force about the seal **122** encircling the filter media **121**.

[0035] In another example, the filter window **114** defines: a window width (e.g., between 3 inches and 4 inches); and a window height (e.g., between 0.5 and 1.0 inches) less than the window width. In this example, the filter assembly **120**: defines a semi-elliptical geometry matching the window width and the window height; and is configured to laterally and vertically deform responsive to an applied force about the periphery of the filter assembly **120**. Accordingly, the user can then: apply a force about the seal **122** encircling the filter media **121** to operate the filter assembly **120** in a deformed configuration to locate the filter assembly **120** within the window width and the window height of the filter window **114**; and remove the applied force about the seal **122** to operate the filter assembly **120** in the nominal configuration to mate the center groove **123** with the rim of the filter window **114**, thereby forming the interference fit between the filter assembly **120** and the filter window **114**.

[0036] In one implementation, the filter assembly **120** can be interchanged across multiple filter

windows located across the shield center section **110**. For example, the shield center section **110** can include: a left filter window **114** arranged proximal the left side of the shield center section **110**; and a right filter window **116** arranged proximal the right side of the shield center section **110**. Accordingly, the filter assembly **120** is selectively arranged within the left filter window **114** or the right filter window **116** by the user. More specifically, the filter assembly **120**: is operable in a first configuration including the center groove **123** of the seal **122** mating a first rim **115** of the left filter window **114** to transiently retain the filter assembly **120** within the left filter window **114**; and is operable in a second configuration including the center groove **123** of the seal **122** mating a second rim of the right filter window **116**, opposite the left filter window **114**, to transiently retain the filter assembly **120** within the right filter window **116**.

[0037] In another implementation, the respirator apparatus **100** includes a first filter assembly **120** including: a first filter media **121**; and a first seal **122** encircling the first filter media **121** and defining a first center groove **123** configured to transiently couple the left filter window **114** of the shield center section **110**. Similarly, the respirator apparatus **100** can include a second filter assembly **128** including: a second filter media **129**; and a second seal **130** encircling the second filter media **129** and defining a second center groove **131** configured to transiently couple the right filter window **116** of the shield center section **110**. In this implementation, as described above, the first filter assembly **120** and the second filter assembly **128** are interchangeable across the left filter window **114** and the right filter window **116**. Additionally, although this implementation includes a first filter assembly **120** and a second filter assembly **128** configured to transiently couple the shield center section **110**, other variations can include a single filter assembly and/or multiple (e.g., greater than two) filter assemblies configured to transiently couple filter windows arranged across the shield center section **110**.

[0038] Therefore, the filter assembly **120** transiently couples the filter window **114** of the shield center section **110** in order to: during inhalation by the user, prevent contaminants (e.g., airborne particulates, dirt, dust) from entering an interior volume of the shield center section **110**; and form an interference fit between the filter assembly **120** and the filter window **114** to direct airflow from outside the shield center section **110** through the filter assembly **120** toward the interior volume of the shield center section **110**.

4.2 Filter Covers

[0039] In one implementation, the filter assembly **120** includes a seal **122**: encircling the periphery of the filter media **121**; defining a center groove **123** extending about the periphery of the seal **122**; and defining an intermittent outer groove **127**—adjacent the center groove **123**—encircling the periphery of the seal **122** and configured to transiently couple a filter cover **133** extending across the filter media **121** of the filter assembly **120**. In this implementation, the filter cover **133** includes: a pre-filter media **134** (e.g., siliconized, Sil-Scrim) heat molded polyester scrim, carbon scrim); and a capture detent **135** arranged about an internal periphery of the pre-filter media **134** and configured to couple the outer intermittent groove **127** to transiently retain the filter cover **133** across an outer face of the filter media **121**. Accordingly, during retention of the filter cover **133** across the outer face, the filter cover **133** cooperates with the filter assembly **120** to form a plenum between the surface area of the filter media **121** of the filter assembly **120** and the pre-filter media **134** of the filter cover **133**.

[0040] Additionally, the filter cover **133** can include a valve plate **170** configured to mate to the filter rim geometry: centrally located on the pre-filter media **134**; arranged within the plenum formed by the filter assembly **120** and the filter cover **133**; and configured to avoid inward airflow during inhalation by the user and outward airflow during exhalation by the user. Thus, the filter cover **133** cooperates with the filter assembly **120** to isolate the filter media **121** of the filter assembly **120** from an external environment in order to phase airflow (e.g., inward airflow, outward airflow) and prevent large particulates and liquid contaminants (e.g., >PM10 airborne particulates, dirt, dust, body fluids, blood, water droplets) from directly contacting the filter media **121** of the

filter assembly **120**.

[0041] Additionally, the valve plate **170**, centrally located on the pre-filter media **134** suspended proximal to the inlet opening of the filter assembly **120** separated from the filter assembly **120** inlet face by the plenum as to prevent interference with inward airflow during inhalation or outward airflow during exhalation. The valve plate **170** is configured to cover the inlet face of the filter assembly **120**. The configuration of the pre-filter media **134** incorporates flexibility of the outer plane of the pre-filter media **134** that suspends the valve plate **170** proximal to the face of the filter assembly **120**. Upon demand the valve plate **170** can be deflected by force to engage the face of the filter assembly **120** and obstruct inlet and outlet airflow to allow an evaluation of the gasket **140** seal mating integrity to the face of the user. The force to engage the valve plate **170** to the face of the filter assembly **120** is less than 1 inch-pound so as not to induce a false positive gasket seal mating integrity evaluation by adding pressure to the gasket **140** seal during the applied pressure sequence of the evaluation process. The evaluation requires short and abrupt inhalation bursts sufficient to generate a vacuum in the plenum of the respirator while the valve plates **170** are simultaneously deflected and pressed against the faces of the filter assemblies **120**. The resulting increase in vacuum caused by an abrupt inhalation draws the respirator apparatus **100** toward the face increasing pressure from the gasket **140** onto the facial sealing land readily perceived by the user. Weak or lack of vacuum during the evaluation indicates gasket seal leakage at the gasket-face sealing land and requires refitting the respirator apparatus **100** until a perceptible vacuum can be performed.

[0042] In one example, the respirator apparatus **100** includes a first filter assembly **120** including a first seal **122** defining: a first center groove **123** extending about the periphery of the first seal **122**; and a first intermittent outer groove **127**—adjacent the first center groove **123**—encircling the periphery of the first seal **122**. In this example, the first filter assembly **120** can transiently couple the left filter window **114** to locate the first outer groove **127** at an exterior side of the lens center section **110**. Additionally, the respirator apparatus **100** includes a first filter cover **133** including: a first pre-filter media **134**; and a first capture detent **135** arranged about a periphery of the first pre-filter media **134** and configured to couple the first intermittent outer groove **127** to transiently retain the first filter cover **133** across a first outer face of the first filter media **121** of the first filter assembly **120**. Similarly, the respirator apparatus **100** includes a second filter assembly **128** including a second seal **130** defining: a second center groove **131** extending about the periphery of the first seal **122**; and a second intermittent outer groove **132**—adjacent the second center groove **131**—encircling the periphery of the second seal **130**. The second filter assembly **128** can transiently couple the right filter window **116** to locate the second outer groove **132** at the exterior side of the lens center section **110**. Furthermore, the respirator apparatus **100** includes a second filter cover **136** including: a second pre-filter media **137**; and a second capture detent **138** arranged about a periphery of the second pre-filter media **137** and configured to couple the second intermittent outer groove **132** to transiently retain the second filter cover **136** across a second outer face of the second filter media **129** of the second filter assembly **128**.

[0043] Therefore, the respirator apparatus **100** can include a filter cover **133** configured to transiently couple the filter assembly **120** in order to: prevent large particulate and liquid contaminants (e.g., >PM10 airborne particulates, dirt, dust, body fluids, blood, water droplets) from directly contacting the filter media **121** of the filter assembly **120**; and direct airflow (e.g., inward airflow, outward airflow) during inhalation and exhalation by the user.

4.3 Adjustable Filter Depth

[0044] In one implementation, the filter assembly **120**: is operable in a first depth configuration including the center groove **123** of the seal **122** coupling the filter window **114** to locate the filter assembly **120** at a first depth (e.g., between 0.1 and 0.2 inches) within the interior volume of the lens center section **110**; and is operable in a second depth configuration including the outer groove **127** of the seal **122** coupling the filter window **114** to locate the filter assembly **120** at a second

depth, greater than the first depth (e.g., between 0.21 and 0.3 inches), within the interior volume of the lens center section **110**. More specifically, the filter assembly **120**: is operable in the first depth configuration including the center groove **123** of the seal **122** mating with the rim of the filter window **114** to transiently retain the filter assembly **120** within the filter window **114** at the first depth; and is operable in the second depth configuration including the outer groove **127** of the seal **122** mating with the rim of the filter window **114** to transiently retain the filter assembly **120** within the filter window **114** at the second depth greater than the first depth.

[0045] Therefore, the filter assembly **120** is selectively arranged within the filter window **114** to adjust depth of the filter assembly **120** extending within the interior volume of the lens center section **110** in order to prevent direct contact of the filter assembly **120** with the face of the user when larger filters are installed. Although, this implementation includes the seal **122** defining a center groove **123** and an outer groove **127** about the periphery of the seal **122**, other variations can include a single groove and/or multiple (e.g., greater than two) grooves arranged about the periphery of the seal **122** encircling the filter media **121** of the filter assembly **120**.

5. Gasket

[0046] Generally, the respirator apparatus **100** includes a gasket **140**: arranged about a rear side of the lens center section **110**; and configured to conform against the face of the user. More specifically, the gasket **140**: is arranged (e.g., adhesively coupled, overmolded) about a periphery of the lens center section **110**; is configured to contour about the face of the user to locate the nose and mouth of the user within the transparent center region **112** of the shield center section **110**; and forms an airflow channel **150** within the interior volume of the shield center section **110** to direct airflow—during inhalation and exhalation by the user—across a glabrous cheek region of the user. Thus, the gasket **140**: during movement of the face by the user, maintains contact against the face of the user to prevent air from entering and/or exiting the interior volume (e.g., plenum) of the lens center section **110** along the gasket **140** compressed against the face of the user; and induces a passive cooling effect—during inhalation and exhalation by the user—across the face of the user as air passing through the airflow channel **150** flows across the glabrous cheek region of the user.

5.1 Facial Compression

[0047] In one implementation, the gasket **140**: is formed of an elastomeric closed cell conformal foam material including a non-porous surface (e.g., integrated skin); is arranged about the rear side of the shield center section **110**; and is configured to conform against the face of the user to locate the filter window **114** proximal the glabrous cheek region of the user. In particular, the gasket **140**: is formed of a elastomeric urethane foam and/or a elastomeric silicone foam material; includes a continuous smooth skin surface integrally formed during the specialized foam molding process that implements a proprietary tool coating that breaks surface tension between the tool contact surfaces and the foam, on contact, collapsing the air cells near the surface of the foam, forming a uniformly smooth skin surface that has increased durability, improved release characteristics and an ideal sealing land surface finish against facial contours; and defines a continuous boundary about the rear side of the lens center section **110** to contour the nose, cheeks, and chin of the user to maintain contact against the face of the user during movement of the face (e.g., movement of the mandible) by the user.

[0048] In this implementation, the gasket **140** includes a left gasket section **141**: defining a first depth (e.g., 0.25 inches, 6 millimeters) contoured along a left edge of the lens center section **110**; and configured to conform a left side of a mandible of the user to locate the left filter window **114** of the lens center section **110** proximal a left glabrous cheek region of the user. Similarly, the gasket **140** includes a right gasket section **142**: matching the first depth (e.g., 0.20 inches, 5 millimeters) contoured along a right edge of the lens center section **110**; and configured to conform against the right side of the mandible of the user to locate the right filter window **116** of the lens center section **110** proximal a right glabrous cheek region of the user. The entire gasket **140** is a single continuous injection molded part designed to eliminate all mold parting lines that create

potential leak paths between the sealing land of the gasket and facial topology.

[0049] Accordingly, the right gasket section **142** cooperates with the left gasket section **141** to: maintain contact between the gasket **140** and the face of the user (e.g. Sealing Land); and direct airflow through the filter assembly **120** into the interior volume of the lens center section **110** (e.g. Plenum). Thus, during movement of the face by the user (e.g., movement of the mandible, turning of the head), the gasket **140**—by the design of the gasket's elastomeric, foam conforming characteristics and geometry sculpted specifically to move with the gasket sealing land's interface on facial topology—maintains contact with the face of the user.

[0050] Additionally, the gasket **140** includes an upper gasket section **143**: defining a second depth greater than the first depth (e.g., 0.25 inches, 6 millimeters); defining an arched geometry contoured along an upper edge of the transparent center region **112**; and configured to conform against the skin supported by a nasal bone and a zygomatic bone of the user to locate the nose of the user within the transparent center region **112** of the shield center section **110**. Furthermore, the gasket **140** includes a lower gasket section **144**: defining third depth greater than the second depth (e.g., 0.80 inches, 20 millimeters); defining a concave geometry contoured along a lower edge of the transparent center region **112**; and configured to conform against a bottom side of the mandible of the user to locate a chin and the mouth of the user within the transparent center region **112** of the lens center section **110**.

[0051] Accordingly, the lower gasket section **144** cooperates with the upper gasket section **143**, the left gasket section **141**, and the right gasket section **142** in order to maintain contact against the skin supported by facial bones (e.g., nasal bone, zygomatic bone, mandible) of the user while maintaining glabrous cheek regions exposed within an interior volume of the shield center section **110**. Therefore, rather than the gasket **140** sliding across the face of the user during movement of the face (e.g., head turns, mandible movement during speaking) by the user, the gasket **140** can elastically deflect, thereby: maintaining contact between the upper gasket section **143** and the nasal bone of the user; and maintaining contact between the lower gasket section **144** and the bottom side of the mandible of the user.

5.2 Airflow Channel+Facial Cooling

[0052] In one implementation, the gasket **140** is configured to conform against the face of the user to: locate the filter assembly **120** offset a glabrous cheek region of the user; and form an airflow channel **150** extending from the filter assembly **120** to the transparent center region **112**, locating the nose and the mouth of the user, and configured to direct airflow across the glabrous cheek region of the user. In particular, the airflow channel **150**: defines a target depth range (e.g., between 0.1 inches and 0.3 inches, between 2.5 millimeters and 8 millimeters) between an interior surface of the shield center section **110** and the face of the user; and defines a target height range (e.g., between 0.5 inches and 1.5 inches, between 2.5 millimeters and 8 millimeters) across the glabrous cheek region of the user.

[0053] In one example, the gasket **140** is configured to conform against the face of the user to: locate the first filter assembly **120** offset a left glabrous cheek region of the user; and form a left airflow channel **150** extending from a first filter assembly **120** at the left filter window **114** to the transparent center region **112** of the shield center section **110** and configured to direct airflow—flowing within the left airflow channel **150**—across the left glabrous cheek region of the user. Similarly, the gasket **140** is configured to compress against the face of the user to: locate the second filter assembly **128** offset a right glabrous cheek region of the user; and form a right airflow channel **150** extending from the second filter assembly **128** at the right filter window **116** to the transparent center region **112** of the shield center section **110** and configured to direct airflow—flowing within the right airflow channel **150**—across the right glabrous cheek region of the user. Accordingly, air travels within the left airflow channel **150** across the left glabrous cheek region of the user and the right airflow channel **150** across the right glabrous cheek region of the user, thereby inducing a cooling effect for the user during inhalation and exhalation by the user.

[0054] For example, during inhalation by the user, the first filter assembly **120** cooperates with the second filter assembly **128** to: direct airflow within the left airflow channel **150**—flowing from the first filter assembly **120** to the mouth of the user—laterally and vertically across the left glabrous cheek region; and direct airflow within the right airflow channel **150**—flowing from the second filter assembly **128** to the mouth of the user—laterally and vertically across the right glabrous cheek region. Additionally, during exhalation by the user, the first filter assembly **120** cooperates with the second filter assembly **128** to: direct airflow within the left airflow channel **150**—flowing from the mouth of the user to the first filter assembly **120**—laterally and vertically across the right glabrous cheek region; and direct airflow within the right airflow channel **150**—flowing from the mouth of the user to the second filter assembly **128**—laterally and vertically across the right glabrous cheek region.

[0055] In another example, during inhalation by the user, the first filter assembly **120** cooperates with the second filter assembly **128** to: direct airflow within the left airflow channel **150**—flowing from the first filter assembly **120** to the nose of the user—laterally and vertically across the left glabrous cheek region; and direct airflow within the right airflow channel **150**—flowing from the second filter assembly **128** to the nose of the user—laterally and vertically across the right glabrous cheek region. Additionally, during exhalation by the user, the first filter assembly **120** cooperates with the second filter assembly **128** to: direct airflow within the left airflow channel **150**—flowing from the nose of the user to the first filter assembly **120**—laterally and vertically across the right glabrous cheek region; and direct airflow within the right airflow channel **150**—flowing from the nose of the user to the second filter assembly **128**—laterally and vertically across the right glabrous cheek region.

[0056] Therefore, the gasket **140** is configured to conform against the face of the user to form the airflow channel **150** within the interior volume of the shield center section **110** in order to induce a cooling effect across the glabrous cheek region of the user during inhalation (e.g., nasal inhalation, oral inhalation) and exhalation (e.g., nasal exhalation, oral exhalation) by the user.

5.3 Moisture Control

[0057] In one implementation, the shield **110**: is formed of a transparent flexible material (e.g., polycarbonate film); and includes a hydrophilic coating arranged across an interior side and an exterior side of the lens center section **110**. Inhalation and exhalation by the user results in formation of moisture (e.g., fog, droplets) across the interior surface of the lens center section **110** which in turn obstructs facial cues (e.g., expressions) made by the user. Accordingly, the hydrophilic coating across the lens center section **110**: prevents formation of moisture (e.g., fog, droplets) across the interior surface of the interior section; and directs moisture—by way of gravity—to lower gasket section **144** of the gasket **140** arranged about the shield center section **110**.

[0058] In another implementation, the gasket **140** can include a moisture trap **145** (e.g. cavity depression) molded into the lower section and cooperating with the hydrophilic coating and moisture shedding gasket skin surface to capture moisture formed within the interior side of the lens center section **110**. Thus, the gasket **140** enables the user to routinely (i.e., as needed) dispose of moisture collected within the moisture trap **145** molded into the lower gasket section **144** of the gasket **140** by the user manually breaking the gasket seal under and adjacent the trap area during any exhalation phase of the breathing cycle so as not to contaminate the plenum space with particulate. Other variations of this implementation can include a one-way release valve inlet at the base of the moisture trap **145** arranged at the lower gasket section **144** and configured to selectively release moisture accumulated within the moisture trap **145** at the lower gasket section **144** by the user performing a user seal check forcing the moisture trap valve to open providing a path for moisture collected in the trap to escape the respirator so as not to contaminate the plenum space with particulate.

6. Headband

[0059] Generally, the respirator apparatus **100** can include: a neck harness **160** coupled to the shield

center section **110** and configured to wrap around a neck of the user; and a cranial harness **165** coupled to the shield center section **110** and configured to wrap about an upper cranium of the user. In particular, the cranial harness **165** cooperates with the neck harness **160** to: locate the shield center section **110** proximal the face of the user; and maintain conformance of the gasket **140** against the face of the user. Thus, during movement of the face by the user, the neck harness **160** cooperates with the cranial harness **165** to, rather than slide across the face of the user, maintain conformance of the gasket **140** against the face of the user.

[0060] In one implementation, the shield center section **110** includes: a lower left perforation **118** arranged proximal the left filter window **114** of the shield center section **110**; and a lower right perforation **118** arranged proximal the right filter window **116** of the shield center section **110** opposite the lower left perforation **118**. Additionally, the neck harness **160** includes: a first harness clip **162** (e.g., snigger clip) arranged at a first end of the neck harness **160** and transiently coupled to the lower left perforation **118** of the lens center section **110**; and a second harness clip **162** (e.g., snigger clip) arranged at a second end, opposite the first end, of the neck harness **160** and transiently coupled to the lower right perforation **118** of the lens center section **110**.

[0061] In one example, the neck harness **160**: is formed of an elastomeric material (e.g., laminated foam backed microfiber, woven polyester flat webbing); and spans a target length (e.g., 26 inches, 66.04 centimeters) from the first end to the second end of the neck harness **160**; and spans a target height (e.g., 0.5 inches, 12 millimeters) configured to contour along a rear neck of the user.

[0062] Additionally, the shield center section **110** includes: an upper left perforation **118** arranged proximal the left filter window **114** of the shield center section **110**; and an upper right perforation **118** arranged proximal the right filter window **116** of the shield center section **110** opposite the upper left perforation **118**. Furthermore, the cranial harness **165** includes: a rear cranial section **166**—formed of the elastomeric material as described above configured to wrap about a rear cranium of the user; a third harness clip **162** (e.g., snigger clip) arranged at a first end of the rear cranial section **166** and transiently coupled to the upper left perforation **118** of the shield center section **110**; and a fourth harness clip **162** (e.g., snigger clip) arranged at a second end, opposite the first end, of the rear cranial section **166** and transiently coupled to the upper right perforation **118** of the shield center section **110**. The cranial harness **165** also includes a front cranial section **167** (e.g., thermoplastic strip): coupled the rear elastomeric cranial section **166**; and including an extended dogleg section oriented rearward on either side to wrap about a front cranium of the user; and cooperating with the rear cranial section to form a halo encircling an upper cranium of the user. This cranial harness provides the additional utility of setting the adjustment of the yoke for one hand respirator donning to the face of the user.

[0063] Accordingly, the cranial harness **165** cooperates with the neck harness **160** to locate the cranial harness **165** taut about the upper cranium of the user providing pressure to stabilize the gasket **140** against the face of the user. Therefore, during movement of the face by the user, the neck harness **160** cooperates with the neck harness **160** to maintain conformally the gasket **140** against the face of the user.

7. Respirator Setup+Facial Configuration

[0064] In one implementation, the respirator apparatus **100** includes the shield center section **110** including: the gasket **140** arranged about the rear side of the shield center section **110**; the transparent center region **112** configured to locate over the face of the user; the left filter window **114** laterally offset from the transparent center region **112**; and the right filter window **116** laterally offset from the transparent center region **112** opposite the left filter window **114**. During initial setup of the respirator apparatus **100**, the user can: couple a first filter assembly **120** within the left filter window **114** of the shield center section **110**; couple the second filter assembly **128** within the right filter window **116** of the shield center section **110**; couple the neck harness **160** to the shield center section **110**; and couple the cranial harness **165** to the shield center section **110**.

[0065] The neck harness **160** is configured to support the respirator apparatus **100** about the rear

neck of the user and maintain the respirator apparatus **100** in a suspended configuration about the user's neck prior to placing the gasket **140** about the face of the user. The user can then: conform the gasket about the face of the user to induce direction of airflow through the filter assembly **120** into the interior volume of the shield center section **110**; and secure the cranial harness **165** in a halo configuration about the upper cranium of the user. Accordingly, the cranial harness **165** cooperates with the neck harness **160** to maintain conformally the gasket **140** against the face of the user creating a sealed plenum between the shield and the face of the user.

[0066] In the sealed configuration, the gasket **140** maintains conformally against the face of the user to form the left airflow channel **150** across the left glabrous cheek region of the user and the right airflow channel **150** across the right glabrous cheek region of the user. Accordingly, in the sealed configuration, the first filter assembly **120** is located offset the left glabrous cheek region of the user and the second filter assembly **128** is located offset the right glabrous cheek region of the user. Thus, during inhalation (e.g., nasal inhalation, oral inhalation) and exhalation (e.g., nasal exhalation, oral exhalation) the respirator apparatus **100** induces a cooling effect across the left glabrous cheek region and the right glabrous cheek region of the user.

[0067] Additionally, the gasket **140** cooperates with the face of the user such that—during movement of the face by the user—rather than slide across the face of the user, the gasket **140** elastically deforms while maintaining contact with the face of the user while directing airflow within the left airflow channel **150** and the right airflow channel **150**.

[0068] Therefore, the respirator apparatus **100**: induces a cooling effect across the glabrous cheek region of the user during inhalation and exhalation of the user; enables an outside observer to view facial cues from the nose and the mouth region of the user located within the transparent center region **112** of the shield center section **110**; and prevents contaminants (e.g., airborne particulates, dirt, dust) about an exterior environment of the lens center section **110** from entering the interior of the lens (e.g., plenum) center section **110**.

[0069] As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the embodiments of the invention without departing from the scope of this invention as defined in the following claims.

Claims

1. A respirator apparatus comprising: a shield: configured to locate over a face of a user; and defining a first filter window; a first filter assembly comprising: a first filter media; and a first seal: arranged about the first filter media; and defining a first circumferential groove configured to mate with the first filter window to: seal a first sealing surface of the first circumferential groove against an exterior surface of the first filter window; seal a second sealing surface of the first circumferential groove against an interior surface, opposite the exterior surface, of the first filter window; and transiently retain the first filter assembly within the first filter window; and a gasket coupled to the shield and configured to: conform against the face of the user; and locate the first filter assembly offset from the face of the user to form a first airflow channel between the first filter assembly and the face of the user.

2. The respirator apparatus of claim 1: wherein the shield defines: a transparent center region; and the first filter window offset from the transparent center region; and wherein the gasket is configured to: conform against the face of the user to align the center region with a nose and a mouth of the user; and locate the first filter assembly offset from a first cheek region of the user to form the first airflow channel.

3. The respirator apparatus of claim 1: wherein the first seal further comprises a first outer groove parallel to and offset from the first circumferential groove; and wherein the first filter assembly is operable: in a first configuration with the first circumferential groove mated to the first filter window to transiently retain the first filter assembly within the first filter window at a first depth

from the face of the user; and in a second configuration with the first outer groove mated to the first filter window to transiently retain the first filter assembly within the first filter window at a second depth, less than the first depth, from the face of the user.

4. The respirator apparatus of claim 3: wherein the shield further defines a second filter window offset from the first filter window; further comprising a second filter assembly: comprising: a second filter media; and a second seal: arranged about the second filter media; defining a second circumferential groove; and a second outer groove offset from the second circumferential groove; operable in a third configuration to transiently retain the second filter assembly within the second filter window by mating the second circumferential groove with a second rim of the second filter window; and operable in a fourth configuration to transiently retain the second filter assembly within the second filter window by mating the second outer groove with the second rim of the second filter window; and wherein the gasket is further configured to locate the second filter assembly offset from a second check region of the user to form a second airflow channel.

5. The respirator apparatus of claim 1: wherein the first seal further comprises a first outer groove parallel to and offset from the first circumferential groove; and further comprising a filter cover comprising: a pre-filter media comprising a first detent configured to couple to the first outer groove of the first seal to: transiently retain the filter cover across an outer face of the first filter media; and form a plenum between the outer face of the first filter media and an inner face of the pre-filter media.

6. The respirator apparatus of claim 1, wherein the first circumferential groove: comprises: the first sealing surface; the second sealing surface arranged opposite the first sealing surface; and a channel interposed between the first sealing surface and the second sealing surface; and is configured to transiently couple to the first filter window to: locate a first rim of the first filter window of the shield within the channel; seal the first sealing surface against the exterior surface of the first rim of the first filter window; and seal the second sealing surface against the interior surface of the first rim of the first filter window, the second sealing surface cooperating with the first sealing surface to form an interference fit against the exterior surface and the interior surface of the first rim of the first filter window.

7. The respirator apparatus of claim 1, wherein the first filter assembly is configured to: during inhalation by the user: direct air, flowing through the first filter media, laterally and vertically across the face of the user toward an orifice of the user; and during exhalation by the user: direct air, exiting the orifice of the user, laterally and vertically across the face of the user toward the first filter media.

8. The respirator apparatus of claim 7: wherein the shield further defines a second filter window laterally offset from the center region opposite the first filter window; further comprising a second filter assembly comprising: a second filter media; and a second seal: arranged about the second filter media; and defining a second circumferential groove configured to mate with the second filter window to transiently retain the second filter assembly within the second filter window; and wherein the gasket is further configured to locate the second filter assembly offset from a second check region of the user to form a second airflow channel between the second filter assembly and the second check region of the user.

9. The respirator apparatus of claim 1, wherein the first filter assembly is configured to: laterally and vertically deform about the first filter window during installation of the first filter assembly into the first filter window and mating of the first circumferential groove against the first filter window; and seal the first circumferential groove against a first rim of the first filter window following installation of the first filter assembly into the first filter window.

10. The respirator apparatus of claim 1: wherein the shield further defines: a set of lower perforations; and a set of upper perforations; and further comprising: a neck harness: comprising a first set of clips configured to couple to the set of lower perforations; and configured to wrap about a rear neck of the user; and a cranial harness: comprising a second set of clips configured to couple

to the set of upper perforations; configured to wrap about a cranium of the user; and configured to cooperate with the neck harness to locate the gasket conformally to the face of the user to form the first airflow channel.

11. The respirator apparatus of claim 1, wherein the gasket: comprises an elastomeric material; and is configured to conformally mate a sealing land to expose glabrous skin regions of the user and direct airflow from outside the shield through the first filter assembly across the glabrous skin regions into an internal volume of a shield plenum.

12. The respirator apparatus of claim 1, wherein the gasket comprises: a first gasket section: arranged along a first lateral edge of the shield; and configured to conform to a first side of a mandible of the user to locate the first filter window proximal a glabrous skin region of the user; and a second gasket section: arranged along a second lateral edge, opposite the first lateral edge, of the shield; configured to conform to a second side, opposite the first side, of the mandible of the user; and cooperating with the first gasket section to, during movement of the mandible by the user, maintain contact between the gasket and the face of the user.

13. The respirator apparatus of claim 12, wherein the gasket further comprises: an upper gasket section: arranged along an upper edge of a center region of the shield; and configured to conform against skin supported by a nasal bone of the user to locate the nose of the user within the center region of the shield; and a lower gasket region: arranged along a lower edge of the center region of the shield; configured to conform to a lower side of the mandible of the user to locate a chin and the mouth of the user within the center region of the shield; and cooperating with the upper gasket section to elastically deflect, during movement of the mandible by the user, to: maintain contact between the upper gasket section and the skin on the nasal bone area of the user; and maintain contact between the lower gasket section and the skin on the lower side of the mandible of the user.

14. The respirator apparatus of claim 1: wherein the shield comprises a hydrophilic coating arranged across an interior side of the shield; and wherein the gasket comprises a moisture trap: molded into a lower section of the gasket; and cooperating with the hydrophilic coating to capture moisture formed within the interior side of the shield.

15. A respirator apparatus comprising: a shield defining a filter window; a filter assembly: comprising: a filter media; and a seal: arranged about the filter media; defining a circumferential groove; and defining an outer groove offset from the circumferential groove; and operable in: a first configuration to transiently retain the filter assembly within the filter window by mating the circumferential groove to a rim of the filter window; and a second configuration to transiently retain the filter assembly within the filter window by mating the outer groove to the rim of the filter window; and a gasket coupled to the shield and configured to conform against a face of the user to locate the filter assembly offset from the face of the user.

16. The respirator apparatus of claim 15: wherein, in the first configuration, the circumferential groove mates with the rim of the filter window to transiently retain the filter assembly at a first depth from the face of the user; and wherein, in the second configuration, the outer groove mates with the rim of the filter window to transiently retain the filter assembly a second depth, less than the first depth, from the face of the user.

17. The respirator apparatus of claim 15, wherein the circumferential groove: comprises: a first sealing surface; a second sealing surface arranged opposite the first sealing surface; and a channel interposed between the first sealing surface and the second sealing surface; and is configured to, in the first configuration, transiently couple to the filter window to: locate a rim of the filter window within the channel; seal the first sealing surface against an exterior surface of the rim of the filter window; and seal the second sealing surface against an interior surface, opposite the exterior surface, of the rim of the filter window, the second sealing surface cooperating with the first sealing surface to form an interference fit against the exterior surface and the interior surface of the rim of the filter window.

18. The respirator apparatus of claim 15, further comprising a filter cover comprising: a pre-filter

media comprising a detent configured to couple to the outer groove of the seal to: transiently retain the filter cover across an outer face of the filter media; and form a plenum between the outer face of the filter media and an inner face of the pre-filter media.

19. A respirator apparatus comprising: a shield defining a filter window; a filter assembly comprising: a filter media; and a seal: arranged about the filter media; and defining a circumferential groove configured to mate with the filter window to: seal opposing sealing surfaces of the circumferential groove against exterior surfaces and interior surfaces of a rim of the filter window; locate a rim of the filter window within a channel interposed between the opposing sealing surfaces; and transiently retain the filter assembly within the filter window; and a gasket coupled to the shield and configured to conform against the face of the user to locate the filter assembly offset from a face of the user.

20. The system of claim 19: wherein the seal further comprises an outer groove offset from the circumferential groove; and wherein the filter assembly is operable: in a first configuration to transiently retain the filter assembly within the filter window by mating the circumferential groove with the rim of the filter window; and in a second configuration to transiently retain the filter assembly within the filter window by mating the outer groove with the rim of the filter window.
